



▲ STEAM RISING FROM THE VALLEY OF GEYSERS

Asia

The Siberian Traps

Location Northwest Siberia, Russia

The traps (steplike hills typical of a basalt landscape) are the legacy of the largest known volcanic eruption in Earth’s history. About 250 million years ago, numerous fissures and vents in western Siberia released an estimated 720,000 cubic miles (3 million cubic km) of flood basalts over a vast area. A mantle plume may have been responsible for the eruptions, but this is still debated by scientists. The greatest mass extinction in history followed, triggered by dramatic changes in the composition of the atmosphere.

The Valley of Geysers

Location The Kamchatka Peninsula in eastern Siberia, Russia

The world’s second-biggest concentration of geysers is in the valley of the fast-flowing Geysernaya River, which carries geothermal waters from a nearby stratovolcano. The valley contains at least 20 large geysers, the most spectacular of which, Velikan, can send out jets of water 130 ft (40 m) high. Some gush every few minutes, others only once every three hours or more. A massive mudslide in 2007 inundated the valley and shut off many of the geysers, but new ones have become active since.

Azerbaijan Mud Volcanoes

Location The eastern Caucasus and Caspian coast, Azerbaijan

More than half the world’s known mud volcanoes, possibly more than 400, are located in Azerbaijan. Mud volcanoes, also known as sedimentary volcanoes, occur where groundwater at up to 212°F (100°C) mixes with mineral deposits and is forced upward. Reaching the surface through fissures, the mud-water mixture forms a small cone, usually no more than 12 ft (4 m) high. Some mud volcanoes also emit methane, as shown when the Lok-Batan mud cone spectacularly erupted in flames in 1977 and 2001.

The Arabian Highlands

Location Yemen and southwest Saudi Arabia, Southwest Asia

At 12,028 ft (3,666 m), the rocky peak of Jebal Nabi Shu’ayb in the southern sector of this range is the highest point on the Arabian Peninsula. To the west, running parallel to the Red Sea as far north as Makkah (Mecca), an escarpment rises from the coastal plain of Tihamah. To the east, a high plateau slopes toward the desert interior of Saudi Arabia. In contrast, there are areas of moist deciduous forest, high-elevation agricultural terraces, and sparkling mountain streams within the highlands.